

Majid Ghasemi

Ph.D. Student, Electrical & Computer Engineering, University of Waterloo

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RESEARCH INTERESTS

Social and multi-agent reinforcement learning, machine ethics, AI alignment. My work asks how ethical behaviour can be learned and transmitted between agents rather than hard-coded as a fixed reward.

EDUCATION

University of Waterloo

Waterloo, ON, Canada

Ph.D., Electrical & Computer Engineering

Sep 2025 – Present

Advisor: Mark Crowley

Focus: Ethical Decision Making in Reinforcement Learning

Wilfrid Laurier University

Waterloo, ON, Canada

M.A.Sc., Applied Computing (GPA A+)

Sep 2023 – May 2025

Advisor: Dariush Ebrahimi

Thesis: *Reinforcement Learning for Scheduling and Routing: Electric Vehicle Charging and Patrol Scheduling*

Imam Khomeini International University

Qazvin, Iran

B.Sc., Software Engineering (GPA A–)

Sep 2017 – Mar 2022

Advisor: Mahdi Bahaghighat Thesis: Machine Learning for Phishing Websites Detection

PUBLICATIONS

Journal Articles

- [J1] **Efficient Vehicle Patrol Scheduling for Urban Safety: Optimization and Heuristic Approaches.**
Majid Ghasemi, Ibrahim Sorkhoh, Fadi Alzhouri, Dariush Ebrahimi. *Future Generation Computer Systems*, 2026.
- [J2] **Reinforcement Learning and Model-based Planning in Practice: A Survey of Algorithmic Rationale and Domain Applications.**
Majid Ghasemi, Amirhossein Mousavi, Dariush Ebrahimi. *Artificial Intelligence Review*, 2026.
- [J3] **Real-Time Electric Vehicle Routing and Charging Optimization with Battery Life Consideration.**
Dariush Ebrahimi, Majid Ghasemi, Mohammad Kashefi, Fadi Alzhouri. *IEEE Internet of Things Journal*, 2026.
- [J4] **Citation Count Prediction Based on Google Scholar Profiles and Clarivate's Journal Citation Reports.**
Mahdi Bahaghighat, Leila Akbari, Majid Ghasemi, Qin Xin. *Information Research*, 2023.
- [J5] **A High-Accuracy Phishing Website Detection Method Based on Machine Learning.**
Mahdi Bahaghighat, Majid Ghasemi, Figen Ozen. *Journal of Information Security and Applications*, vol. 77, 2023.

Peer-Reviewed Conference Papers

- [C1] **The Forager’s Dilemma: Peer Enforcement Alone Is Not Enough for Stable Multi-Agent Alignment.**
Majid Ghasemi, Prabhjyot Singh, Mark Crowley. *Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.
- [C2] **AI Alignment without Representation: From Scalar Rewards to Social Deliberation.**
Majid Ghasemi, Mark Crowley. *Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.
- [C3] **Scaling Limits of Deep Reinforcement Learning: A Stability Analysis with Maximal Update Parametrization.**
Majid Ghasemi, Mark Crowley. *The 39th Canadian Conference on Artificial Intelligence*, 2026.
- [C4] **Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response.**
Majid Ghasemi, Dariush Ebrahimi. *IECON 2025, 51st Annual Conference of the IEEE Industrial Electronics Society*, 2025.
- [C5] **Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety.**
Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, Dariush Ebrahimi. *IEEE 34th International Symposium on Industrial Electronics (ISIE)*, 2025.
- [C6] **Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging.**
Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *21st International Conference on the Design of Reliable Communication Networks (DRCN)*, 2025.
- [C7] **Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra’s Algorithm.**
Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *International Conference on the Dynamics of Information Systems*, 2024.

Peer-Reviewed Workshop Papers

- [W1] **Learning When to Trust in Contextual Social Bandits.**
Majid Ghasemi, Mark Crowley. *19th European Workshop on Reinforcement Learning (EWRL)*, 2026.
- [W2] **Rethinking AI Alignment: From Static Rewards to Social Reinforcement Learning.**
Majid Ghasemi, Mark Crowley. *Pluralistic Alignment Workshop, ICML*, 2026.
- [W3] **Can Standard MARL Metrics Distinguish Communicative from Strategic Action?**
Majid Ghasemi, Mark Crowley. *PhilML Workshop, ICML*, 2026.
- [W4] **Toward Virtuous Reinforcement Learning: A Critique and Roadmap.**
Majid Ghasemi, Mark Crowley. *Machine Ethics Workshop (From Formal Methods to Emergent Machine Ethics)*, AAAI, 2026.

Preprints and Technical Reports

- [P1] **Objective Decoupling in Social Reinforcement Learning: Recovering Ground Truth from Sycophantic Majorities.**
Majid Ghasemi, Mark Crowley. *arXiv preprint*, 2026.
- [P2] **Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges.**

Majid Ghasemi, Amirhossein Mousavi, Dariush Ebrahimi. *arXiv preprint*, 2024.

[P3] **Introduction to Reinforcement Learning.**

Majid Ghasemi, Dariush Ebrahimi. *arXiv preprint*, 2024.

INDUSTRY EXPERIENCE

Kisoji Biotechnology Inc.

Waterloo, ON, Canada

AI Research Intern, Machine Learning Department

May 2026 – August 2026

- Building the Expressibility Engine, a three-module system for antibody expression optimization.
- Domain-adaptive prediction with Bayesian project embeddings; RL-guided experimental design using GRPO with DPP-based diversity; sparse graph-based causal driver discovery.
- Reduces wet-lab iteration through active learning while producing interpretable guidance for antibody engineers.

HONOURS AND AWARDS

AIES Student Program Scholarship Award	2026
Graduate Student Research and Development Award (×2), University of Waterloo	2026
International Doctoral Student Award, University of Waterloo	2025 – 2029
Academic Excellence Gold Medal, Wilfrid Laurier University	2025
Graduate Scholarship, Wilfrid Laurier University	2023 – 2025
Graduate Fellowship, Wilfrid Laurier University	2023 – 2025
Differential Tuition Fee Waiver, Wilfrid Laurier University	2023 – 2025
Excellence Students Tuition Fee Waiver, Imam Khomeini Intl. University	2017 – 2022

GRANTS

HU University of Applied Sciences Utrecht

Aug 2022 – Aug 2023

Research grant to study the Woman, Life, Freedom movement in Iran using computational social science and machine learning methods.

RESEARCH EXPERIENCE

UWECFML / CompThink Lab, University of Waterloo

Sep 2025 – Present

Graduate Researcher, advised by Mark Crowley

- Developing socially informed and ethically grounded methods for multi-agent reinforcement learning.
- Work spans virtue transmission in sequential social dilemmas, contextual bandits with strategic feedback, and constrained ethical agent design.

Edge Computing Lab, Wilfrid Laurier University

Sep 2023 – May 2025

Graduate Researcher, advised by Dariush Ebrahimi

- Reinforcement learning for intelligent transportation, focusing on EV fleet management, routing, and charging optimization.
- Developed exact and heuristic baselines for urban patrol scheduling under dynamic demand.

AIST Lab, Imam Khomeini International University

Jan 2021 – Mar 2022

Undergraduate Researcher, advised by Mahdi Bahaghighat

- Machine learning methods for phishing website detection.

INVITED TALKS AND PRESENTATIONS

Designing Virtuous Agents with Reinforcement Learning, Formal Ethics 2026, Buffalo, USA	<i>Jul 2026</i>
Optimal and Heuristic Solutions for Vehicle Patrol Scheduling, IEEE ISIE 2025	<i>Jun 2025</i>

TEACHING EXPERIENCE

Teaching Assistant 2021 – Present

- **University of Waterloo:** ECE108 Discrete Mathematics and Logic (S26), ECE406 Algorithm Design and Analysis (W26), ECE250 Data Structures and Algorithms, tutor (F25).
- **Wilfrid Laurier University:** CP468 Artificial Intelligence (F23, F24, W25), CP421 Data Mining (W24), CP312 Algorithm Design and Analysis (W24).
- **Imam Khomeini Intl. University:** Artificial Intelligence, Data Mining, Fundamentals of Programming (2021 – 2022).
- Guest lectured selected topics, authored quizzes and project specifications, designed and evaluated term projects, and held regular office hours.

ACADEMIC SERVICE

Program Committee

- AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026

Reviewer

- Reinforcement Learning in Big Worlds Workshop, Reinforcement Learning Conference (RLC), 2026
- PhilML Workshop, ICML 2026
- Pluralistic Alignment Workshop, ICML 2026
- Post-AGI Science and Society (P-AGI) Workshop, ICLR 2026
- IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)
- ACM Transactions on Intelligent Systems and Technology (TIST)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS

ML frameworks: PyTorch, JAX, TensorFlow

Scientific stack: NumPy, Pandas, Matplotlib

Spoken languages: English (fluent), Persian (native), Azeri (native)